PROFITABILITY & COMMERCIAL PERFORMANCE ANALYTICS

Profitability & Revenue Quality Dashboard

An executive analytics solution to monitor growth, evaluate profitability, and uncover key business drivers for informed commercial decisions.

10.46M
PROFIT
Total profit generated after covering costs

41.97%
PROFIT MARGIN %
Percentage of revenue retained as profit

211.07%
REVENUE GROWTH %
Pevenue growth compared Jan–Jun 2017 vs Jan–Jun 2016

GROWTH ANALYSIS
Understand revenue trends, orders, AOV, and category mix

PROFITABILITY ANALYSIS
Identify profit drivers across products, categories, and regions

RETURNS & CUSTOMERS
Quantify returns impact and analyze customer segments

REGIONAL PERFORMANCE
Evaluate performance across territories and regions

Executive Summary

Adventure Works operates across multiple product categories, customer segments, and sales territories. While revenue growth is a key indicator of commercial success, sustainable business performance depends on understanding whether that growth is translating into profitability.

This project was developed to help a Profitability Analysis Stakeholder evaluate business performance through three lenses: **Revenue Growth**, **Profitability**, and **Revenue Quality** — using transactional sales, returns, customer, product, and territory data to identify revenue drivers, profitability drivers, return-related leakage, customer segment performance, and regional performance, delivered via a two-page Power BI dashboard.

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Business Outcome

The dashboard enables stakeholders to move beyond simple performance monitoring and investigate why business outcomes are changing, supporting more informed commercial decision-making.

Business Context

Revenue growth alone does not guarantee business success. A business can generate strong revenue while simultaneously experiencing the risk factors below — the objective of this project was to determine whether commercial growth was creating sustainable value.

- Margin Erosion
- Rising Costs
- Return-Related Losses
- Product Profitability Issues
- Regional Underperformance

PRIMARY STAKEHOLDER

Profitability Analysis Stakeholder

The objective of this project was to determine whether commercial growth was creating sustainable value and to identify the factors driving both growth and profitability.

STAKEHOLDER RESPONSIBILITIES

- Monitor financial performance
- Evaluate commercial efficiency
- Identify profitability risks
- Allocate resources across products and regions
- Support revenue growth initiatives
- Improve profit margins

Business Questions

The project was structured around eleven business questions designed to support commercial decision-making.

GROWTH ANALYSIS

1. How does monthly revenue and order volume vary across the year, and are seasonal patterns consistent across 2015–2017?
2. What is the revenue contribution by product category, and how has category-level mix shifted over time?
3. What is the AOV trend, and how do AOV and Order Volume contribute to Revenue growth over time?

REGIONAL ANALYSIS

4. Which regions are driving or dragging overall revenue, and has that pattern shifted across the three years?

RETURNS ANALYSIS

5. What is the return rate by product category, does it correlate with order volume, and does it vary significantly across territories?
6. Are high-return products also high-revenue products, and what does that imply for net revenue?

CUSTOMER ANALYSIS

7. Which customer segments generate the highest revenue, and are they growing or shrinking?

PRODUCT & PROFITABILITY ANALYSIS

8. Which products have high volume but low revenue contribution?
9. Which products have high revenue but low profit?
10. Which products, categories, and regions generate the highest profit?
11. Are we growing revenue at the expense of profitability?

KPI Framework

The analytical framework was designed around three outcome KPIs supported by diagnostic and supporting metrics.

OUTCOME KPIS

Profit

Measures total value created after product-related costs are deducted from revenue.

Profit Margin %

Measures how efficiently revenue is converted into profit.

Revenue Growth %

Measures commercial expansion over comparable periods.

DIAGNOSTIC METRICS

These metrics explain why KPI performance changes.

- Revenue
- Cost
- Order Volume
- AOV
- ASP
- ARPC
- Revenue Contribution %
- Return Rate %
- Net Revenue
- Customer Count
- Mix Shift

SUPPORTING METRICS

These metrics provide business context.

- Product Category
- Product
- Region
- Customer Segment
- Month
- Year

Data & Methodology

DATASET

Adventure Works Sales Dataset

DATE RANGE

1 January 2015 – 30 June 2017

DATA SOURCES

- Sales
- Returns
- Customers
- Products
- Product Categories
- Product Subcategories
- Territories
- Calendar

DATA PREPARATION

- Referential integrity
- Grain consistency
- Key uniqueness
- Data type correctness
- Duplicate validation
- Missing value assessment

Data quality assessment and validation were performed against the criteria above to ensure a reliable analytical foundation.

Data Modeling

FACT TABLES

Sales · Returns

DIMENSION TABLES

Customers · Products · Categories · Territories · Calendar

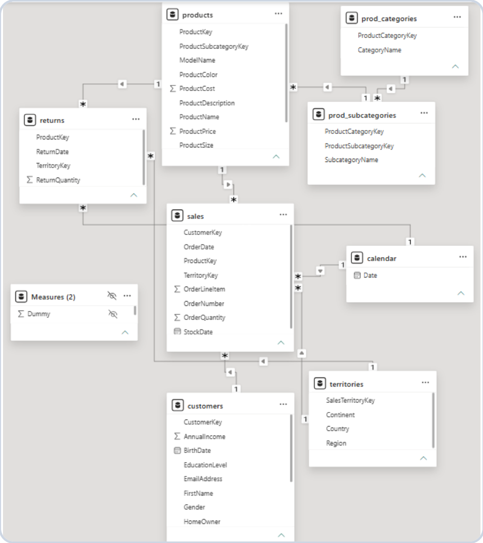


Figure 1. Adventure Works Dimensional Data Model

KEY ASSUMPTIONS

- Revenue derived from Product Price × Order Quantity
- Cost derived from Product Cost × Order Quantity
- Profit calculated as Revenue − Cost
- Dataset contains partial 2017 data (H1 only)
- Returns do not contain return reasons
- Discount information is unavailable

Dashboard Solution

A two-page dashboard was developed to support both executive monitoring and business investigation.

Page 1

Profitability & Revenue Quality Dashboard

Executive Performance & Growth Analysis

DESIGNED TO ANSWER

- How is the business performing?
- Is growth occurring, and what is driving it?
- Which categories contribute most to revenue?

THE PAGE INCLUDES

- KPI monitoring & revenue trend analysis
- Growth diagnostics & revenue contribution analysis
- Profitability trend analysis

Page 2

Business Drivers Dashboard

Deep Dive into Key Drivers of Profitability

DESIGNED TO ANSWER

- What is driving profitability, and where is revenue at risk?
- Which customer segments create value?
- Which regions deserve investment?

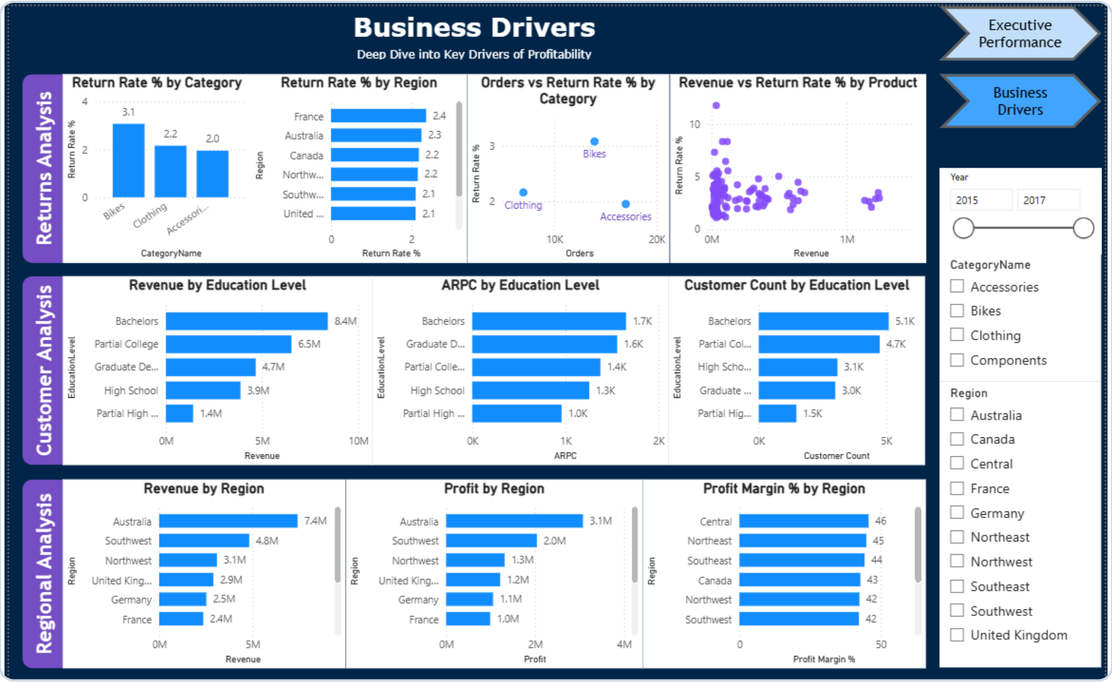
THE PAGE INCLUDES

- Profit decomposition & product profitability analysis
- Returns analysis & customer analysis
- Regional analysis

Dashboard Solution — Page 1

Profitability & Revenue Quality Dashboard — Executive Performance

Dashboard Solution — Page 2



Business Drivers — Deep Dive into Key Drivers of Profitability

Key Insights

Revenue Growth Remains Strong

Revenue Growth reached 211.07%, indicating substantial business expansion, driven primarily by increased commercial activity rather than AOV increases.

Implication: Demand remains strong, and growth initiatives appear effective.

Revenue Is Highly Concentrated

The Bikes category contributes approximately 95% of category revenue.

Risk: Future performance may be vulnerable to category-specific disruptions.

Profitability Remains Healthy

Profit Margin remains near 42%, indicating that growth is generally translating into value creation.

Implication: The business is expanding without obvious large-scale margin erosion.

Returns Create Revenue Leakage

The Bikes category exhibits the highest return rate among major categories.

Implication: The category driving growth also generates the largest return-related losses.

Regional Performance Is Uneven

Australia consistently generates the strongest revenue and profit performance.

Implication: Commercial success is concentrated geographically.

Margin Quality Differs by Category

Accessories generate substantially stronger margins than several larger revenue-generating categories.

Implication: Revenue scale and profitability quality are not always aligned.

Recommendations

Revenue Growth

Reduce dependency on a single category by increasing investment in adjacent product categories with growth potential.

→ **Improved revenue diversification and lower concentration risk**

Profitability Improvement

Review high-revenue, low-profit products to identify pricing, cost, or product-mix opportunities.

→ **Higher profit without requiring proportional revenue growth**

Returns Reduction

Prioritize return reduction initiatives within the Bikes category.

→ **Increased Net Revenue and improved profitability**

Regional Strategy

Analyze commercial practices in Australia and replicate successful approaches in lower-performing territories.

→ **More balanced regional performance**

Product Strategy

Increase cross-selling and attachment opportunities for higher-margin products.

→ **Improved Profit Margin through product mix optimization**

Business Impact Framework & Conclusion

Finding	Recommendation	Expected Business Impact
Revenue concentrated in Bikes	Diversify category growth	Lower revenue concentration risk
High Bike return rate	Return reduction program	Higher Net Revenue
Strong Australian performance	Replicate best practices	Regional growth improvement
Margin variation across categories	Product mix optimization	Higher Profit Margin
High-revenue low-profit products	Profitability review	Improved profit generation

This project transformed transactional business data into a structured profitability analytics framework designed to support commercial decision-making. By combining KPI monitoring, diagnostic analysis, profitability investigation, return analysis, customer segmentation, and regional performance evaluation, the solution enables stakeholders to move from performance observation to actionable decision-making.

The project demonstrates an end-to-end analytics workflow covering business understanding, KPI design, data modeling, data preparation, DAX development, dashboard development, insight generation, and business recommendation formulation.